

Task 20 — Dynamic Dashboarding & Storytelling Capstone Report

Author: Jeevan N | **Track:** Data Analyst | **Environment:** Streamlit / Plotly / Pandas

1. CAPSTONE OBJECTIVE & DASHBOARD ARCHITECTURE

Objective: Synthesize Phase 1 analytics deliverables into an interactive, decision-grade dashboard that combines data exploration with executive storytelling.

Stack: Built using Streamlit and Plotly Express with stateful multi-select filtering across geographic regions and RFM customer segments.

2. DASHBOARD METRIC DICTIONARY

Metric Name	Definition	Data Source	Refresh Frequency
Revenue	Gross monetary value from transactions	Task 19 Pipeline Output	Daily Automated Batch
Orders	Count of fulfilled purchase baskets	Task 18 Basket Log	Daily Automated Batch
Average Order Value	Total Revenue divided by Order Count	Dynamic Aggregation	Real-Time Dynamic
Customer Segment	Behavioral segment mapping	Task 17 RFM Model	Monthly Refresh

3. PITFALL SAFEGUARDS & INTERACTION STABILITY

- **Zero Empty State Errors:** Integrated fallback exception handling so charts degrade gracefully if filters are cleared.
- **Data Freshness & Caching:** Applied `@st.cache_data` wrappers to minimize redundant re-queries and ensure smooth user interaction.
- **Decision Focus:** Placed critical KPI scorecards at top of view to inform immediate executive action.

4. DELIVERABLES & REPOSITORY VERIFICATION

- **Application File:** `app.py`.
- **GitHub Repository:** Live on <https://github.com/JeevanN-07/placemux.git>.